Midterm Project Report

Advanced Computer Programming

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Chapter 1 Introduction

1.1 Github

1) Personal Github Account: https://github.com/110021085

2) GroupProjectRepository:https://github.com/Linx25059/group-5

1.2 Overview

The primary objective of this project is to develop an automated web crawler using the **Scrapy framework** The crawler is designed to precisely extract repository information from a designated GitHub profile. and export the collected data into a structured **XML format** for further analysis.

Chapter 2 Implementation

2.1 Class 1

This class inherits from scrapy.Spider, serving as the central engine of the project. Its primary role is to define the crawling behavior, manage network requests, and parse the HTML structure of GitHub. By leveraging Scrapy's asynchronous processing, it efficiently navigates through repository lists and project details..

2.1.1 Fields

- **name**: Defined as "github_user", which serves as the unique identifier for the crawler.
- **start_urls**: Set to the target user's repository tab: https://github.com/110021085?tab=repositories.
- **custom_settings**: A configuration dictionary that includes a browser-like **USER_AGENT** to bypass bot detection, **FEED_EXPORT_ENCODING** for UTF-8 support, and **ROBOTSTXT_OBEY** set to `False` to ensure full access to the required pages.

2.1.2 Methods

- **parse(self, response)**: The primary entry-point method that iterates through the repository list. It extracts individual repository names and URLs using CSS and XPath selectors, then passes them to the next parsing stage using meta data.
- **parse_details(self, response)**: A specialized callback method that analyzes the internal page of each repository. It performs deep extraction of specific fields such as programming languages and commit counts.

2.1.3 Functions

- **Data Normalization Logic**: Within `parse_details`, the function evaluates if a repository is empty. If the "About" field is missing but the repository contains content, it automatically populates the field with the **Repository Name** to satisfy project requirements.
- **Final Item Yielding**: Packages all scraped data into a structured dictionary (URL, About, LastUpdated, Languages, Number_of_Commits) for XML exportation.

2.2 Class 2

2.3 Method/Function 1

2.4 Method/Function 2

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Chapter 3 Results

3.1 Result 1

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```
2026-04-17 00:43:16 [scrapy.core.engine] INFO: Closing spider (finished)
2026-04-17 00:43:16 [scrapy.extensions.feedexport] INFO: Stored xml feed (2 items) in: midterm_report.xml
2026-04-17 00:43:16 [scrapy.statscollectors] INFO: Dumping Scrapy stats:
{'downloader/request_bytes': 1882,
 'downloader/request_count': 3,
 'downloader/request_method_count/GET': 3,
 'downloader/response_bytes': 147235,
 'downloader/response_count': 3,
 'downloader/response_status_count/200': 3,
 'elapsed_time_seconds': 3.445793,
 'feedexport/success_count/FileFeedStorage': 1,
 'finish_reason': 'finished',
 'finish_time': datetime.datetime(2026, 4, 16, 16, 43, 16, 974266, tzinfo=datetime.timezone.utc),
 'httpcompression/response_bytes': 659713,
 'httpcompression/response_count': 3,
 'item_scraped_count': 2,
 'items_per_minute': 40.0,
 'log_count/DEBUG': 5,
 'log_count/INFO': 3,
 'request_depth_max': 1,
 'response_received_count': 3,
 'responses_per_minute': 60.0,
 'scheduler/dequeued': 3,
 'scheduler/dequeued/memory': 3,
 'scheduler/enqueued': 3,
 'scheduler/enqueued/memory': 3,
 'start_time': datetime.datetime(2026, 4, 16, 16, 43, 13, 528473, tzinfo=datetime.timezone.utc)}
2026-04-17 00:43:16 [scrapy.core.engine] INFO: Spider closed (finished)

C:\Users\USER\github_spider>
```

3.2 Result 2

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
▼<items>
  ▼<item>
      <URL>https://github.com/110021085/110021085</URL>
      <About>None</About>
      <LastUpdated>None</LastUpdated>
      <Languages>None</Languages>
      <Number_of_Commits>None</Number_of_Commits>
    </item>
  ▼<item>
      <URL>https://github.com/110021085/resume-110021085</URL>
      <About>resume-test</About>
      <LastUpdated>None</LastUpdated>
      <Languages>None</Languages>
      <Number_of_Commits>None</Number_of_Commits>
    </item>
  </items>
```

Chapter 4 Conclusions

Through this implementation, I have successfully developed an **automated web crawler** using the **Scrapy framework**. It is capable of automatically navigating to a specific GitHub profile and performing **deep extraction** of each repository's details. Furthermore, I integrated **conditional logic** into the code to handle incomplete web data